

Industrial Vibration Troubleshooting Guide

Synthetic project reference document for Phase 5 RAG testing.

1. Understanding Vibration

Vibration is a useful condition-monitoring signal for rotating machinery. The most useful interpretation comes from trends, operating conditions, and correlation with temperature, RPM, current, and maintenance history.

2. Common Causes

Common causes include imbalance, misalignment, bearing degradation, looseness, coupling problems, tool imbalance, foundation problems, and resonance. Similar symptoms can have different causes.

3. Vibration Trend

A gradual increase may indicate wear or deterioration. A sudden increase may indicate looseness, imbalance, component damage, or a process change. Compare current vibration with the machine's established normal baseline.

4. Correlated Indicators

High vibration plus increasing temperature can strengthen suspicion of bearing or friction-related problems. High vibration plus unstable RPM can indicate rotating-system or drive issues. High vibration plus increasing motor current can indicate increasing mechanical load.

5. Recommended Investigation

Confirm machine state and recent process changes. Review vibration history, temperature, RPM, motor current, recent failures, and maintenance records. Then perform approved physical inspections of bearings, coupling, tool holder, mounting, and alignment.

Important: This is synthetic reference content created for software testing and demonstration. It is not a manufacturer manual or a substitute for approved maintenance procedures, qualified personnel, or applicable safety requirements.